Seat Number: ____

National University of Singapore
Department of Mathematics

Semester 1, AY2026/2027

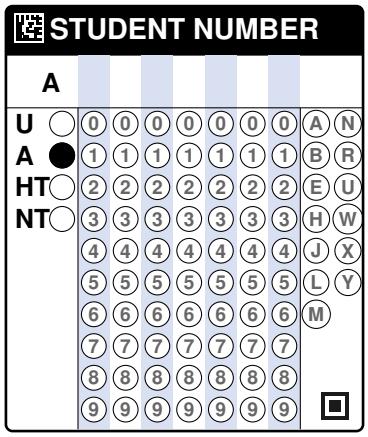
MA1521 Calculus for Computing

Sample Final Exam Paper

October 2026 — Time allowed: 2 hours

INSTRUCTIONS TO CANDIDATES

1. Write your **student number** on the right. For each digit or letter, shade the corresponding circle in the grid **completely** using ink or pencil. DO NOT WRITE YOUR NAME! Do this correctly to get **2 marks**.
2. Write your **seat number** in the box at the upper right hand corner. Do this correctly to get another **2 marks**.
3. This exam paper consists of 8 printed pages (including the cover page). There are 28 questions, of **2 marks each**. Answer ALL questions.
4. This is a closed book exam, but you are allowed to bring in one A4-size and double-sided helpsheet.
5. The use of any form of communication or programmable computing device (e.g. mobile phone, laptop, tablet, etc.) is strictly prohibited during the exam. However, you may use a non-programmable calculator.
6. Write legibly. Follow instructions strictly. You may use a pen (blue or black ink) or a pencil.
7. For each question, only a legible answer written within the given box will be marked. All other writing will be ignored.
8. This booklet – and only this booklet – will be collected at the end of the exam. Do not tear off any page from this booklet. Do not insert any sheet of paper into this booklet.



Total Score: ____

(out of **60 marks**)

1. Evaluate $\displaystyle\int_0^1 x(\ln x)^2\,dx$.

   Give the answer as an integer or a fraction in lowest terms.

2. Evaluate $\displaystyle\lim_{x\to 2}\frac{(x-2)(\ln x-\ln 2)}{x^2-5x+6+2\ln x-\ln 4}$.

   Give the answer as an integer or a fraction in lowest terms.

3. Consider the function f given by

$$f(x)=\int_{x-2}^{x^2-7}\frac{1}{t^3+3}\,dt.$$

   Find $f'(3)$.

   Give the answer as an integer or a fraction in lowest terms.

4. Let $P=(3,2,1)$ and $Q=(0,1,2)$. Find the point of intersection between the plane $2x+3y+6z=14$ and the line through P,Q.

   Give the answer in the form (a,b,c) where a,b,c are real numbers.

5. Let S be the surface in the xyz-space defined by the equation $3x^2-2y+2z^2=4$. Find the point on the surface S at which the tangent plane to S is parallel to the plane $6x-y+2z=5$.

   Give the answer in the form (a,b,c), where a, b, c real numbers.

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6. Find the function $f(x,y)$ such that

$$\frac{\partial f}{\partial x} = \cos x + 4x^3y^2, \quad \frac{\partial f}{\partial y} = 2x^4y - \sin y,$$

and $f(0,0) = 7$.

Give the answer as an expression in terms of x and y.

7. Let $u = 3xy$, $v = v(x,y)$ be such that $\dfrac{\partial v}{\partial x} = 2x$ and $\dfrac{\partial v}{\partial y} = -2y$, $w = x^2 + xy$ and $z = u^2 + 2v + w^2$.

Determine $\dfrac{\partial z}{\partial y}$ as a function of x, y.

8. Consider the function $f(x,y) = x^2y - y^2$. Find the directional derivative of $f(x,y)$ at the point $(3,2)$ in the direction of the vector $8\mathbf{i} + 15\mathbf{j}$.

Give the answer as an integer or a fraction in lowest terms.

9. Let $P(x,y,z)$, measured in pascals, be the air pressure at any point (x,y,z) in the xyz-space. Here the coordinates x, y, z are measured in meters. It is given that $\nabla P(x,y,z) = \langle yz - 3x^2,\ xz,\ xy - 6\rangle$. Suppose the position vector $\mathbf{r}(t)$ of a bird at time t, measured in seconds, is given by $\mathbf{r}(t) = \langle t^2,\ t^3,\ 2t + 1\rangle$. Let A be a real number such that the rate of change of air pressure experienced by the bird at time $t = 1$ is A pascals per second. Find the value of A.

Give the answer as an integer or a fraction in lowest terms.

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10. Determine all saddle points of the function

$$f(x,y) = 2x^3 - 6x + 3xy^2.$$

11. Keep the notation as in the previous question. Write down the system of three polynomial equations in the unkowns x, y and a parameter λ that you will need to solve if you use the method of Lagrange multipliers to find the maximum value of the function $f(x,y) = 2x^3 - 6x + 3xy^2$ on the circle $x^2 + y^2 = 4$.

12. Keep the notation as in the previous question. It is given that $(-1,0)$ is the only local maximum point of the function $f(x,y) = 2x^3 - 6x + 3xy^2$ in the xy-plane. Determine the maximum value of the function $f(x,y)$ on the closed disk of radius 2 centered at $(0,0)$.
Give the answer in the form $a\sqrt{b}$ where both a and b are positive integers and b is as small as possible.

13. Let R be the region in the xy-plane bounded by the lines $x = 0$, $x = 2$, $y = 0$, $y = 4$. Evaluate the double integral $\iint_R (3x^2 + 4xy)\, dA$.
Give the answer as an integer or a fraction in lowest terms.

14. Consider the triangular region R in the xy-plane with vertices at $(1,3)$, $(4,3)$ and $(3,4)$. Compute $\iint_R y^2\, dA$.
Give the answer as an integer or a fraction in lowest terms.

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15. Let R be the region in the xy-plane bounded by the curve $y = x^2$ and the line $y = x + 2$. Find the value of the double integral $\iint_R x\,dA$.

    Give the answer as an integer or a fraction in lowest terms.

16. Find the exact value of the iterated integral $\int_0^1 \int_{x^{1/3}}^1 \frac{1}{\sqrt{y^4+1}}\,dy\,dx$.

17. Let $f(t)$ be a function such that $\int_0^6 f(t)\,dt = 18$. Let R the region in the xy-plane bounded by the lines $x = 0$, $x = 3$, $y = 0$ and $y = 2$. Find the value of the double integral $\iint_R f(2x)f(3y)\,dA$.

    Give the answer as an integer or a fraction in lowest terms.

18. Consider the region R in the first quadrant of the xy-plane bounded by the curve $y = \sqrt{2x - x^2}$ and the x-axis. Find the value of the double integral $\iint_R \frac{1}{x}\,dA$.

    Give the answer in terms of π.

19. Compute $\frac{1}{\pi}\iint_R (3+x^4+2x^2y^2+y^4)\,dA$ where R is the closed disk of unit radius centered at the origin.

    Give the answer as an integer or a fraction in lowest terms.

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20. Let D be the parallelogram in the xy-plane bounded by the lines $y = -x + 5$, $y = -x + 2$, $y = 2x - 1$, $y = 2x - 4$. Compute the integral $\iint_D e^x \, dy dx$.

Give the answer as an expression in terms of integer powers of e.

21. Let R be the region in first quadrant of the xy-plane bounded by the curves $x^2 + y^2 = 9$, $x^2 + y^2 = 16$ and the lines $y = 2x+1$, $y = 2x+2$. Compute the integral $\iint_R (x + 2y)(x^2 + y^2) \, dy dx$.

Give the answer as an integer or a fraction in lowest terms.

22. Consider the double integral

$$\iint_R \frac{y^2}{x} e^{x^2} \, dA$$

where R is the region in the first quadrant of the xy-plane bounded by the curves $x = 1$, $x = \sqrt{3}$, and the lines $y = x$, $y = 2x$. Use the change of variables $u = x^2$ and $v = \dfrac{y}{x}$ to rewrite the above double integral as an iterated integral in the variables u, v and with integration order $du\, dv$.

[You do not have to compute the integral.]

Give the answer in the form $\int_a^b \int_c^d g(u,v) \, du \, dv$, where a and b are real numbers, c and d are real numbers or expressions in terms of v, and $g(u,v)$ is an expression in terms of u and v.

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23. Compute $\int_{-2}^{2}\int_{-2}^{2}\int_{-2}^{2} xz^2\,dzdydx.$

Give the answer as an integer or a fraction in lowest terms.

24. Compute the integral

$$\int_{0}^{1}\int_{0}^{x}\int_{x^2+y^2}^{2} dzdydx.$$

Give the answer as an integer or a fraction in lowest terms.

25. Let R be the solid region in the xyz-space bounded by the surface $x^2+y^2=9$, the surface $x^2+y^2=1$, the plane $z=1$ and the plane $z=0$. Compute the integral

$$\iiint_R z^2\,(x^2+y^2)^2\,dx\,dy\,dz.$$

Give the answer as an expression in terms of π.

26. Consider the solid region E in the xyz-space bounded by the cylinder $x^2+y^2=9$, the horizontal plane $z=2$, and the surface $z=6+x$. Evaluate the triple integral

$$\frac{1}{\pi}\iiint_E (x^2+y^2)\,dV.$$

Give the answer as an integer or a fraction in lowest terms.

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27. Let E be the unit ball consisting of points (x, y, z) in the xyz-space such that $x^2 + y^2 + z^2 \leq 1$. Evaluate the triple integral

$$\iiint_E (x^2 + y^2 + z^2)\, dV.$$

Give the answer in terms of π.

28. Let E be the solid region in the xyz-space consisting of points (x, y, z) such that $\sqrt{x^2 + y^2} \leq z \leq 2$. Rewrite the triple integral

$$\iiint_E \frac{z}{(x^2 + y^2 + z^2 + 2)^2}\, dV$$

as an iterated integral in spherical coordinates and with integration order $d\rho\, d\phi\, d\theta$.

[You do not have to compute the integral.]

Give the answer in the form

$$\int_a^b \int_c^d \int_e^f g(r, \phi, \theta)\, d\rho\, d\phi\, d\theta,$$

where a and b are real numbers, c and d are expressions in terms of θ, e and f are expressions in terms of θ and ϕ, and $g(\rho, \phi, \theta)$ is an expression in terms of ρ, ϕ, θ. Note that some of c, d, e, f may also be real numbers.

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